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Does implied volatility efficiently predict future realized volatility, and how does this relationship vary across market regimes and between cryptocurrency and equity markets?

Gabin Torres & **Clément Massabo**

Thesis Supervisor:

Alexandre Landi

The views expressed in this document are solely those of the authors and should not be considered as investment advice.

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Abstract

This thesis investigates whether implied volatility efficiently predicts future realized volatility, and how this relationship varies across market regimes and between cryptocurrency and equity markets. The analysis focuses on Bitcoin and the S&P 500, using the DVOL and VIX indices as proxies for implied volatility, while realized volatility is constructed from daily log returns.

A unified empirical framework is implemented to allow for direct comparison across markets. Predictive regressions are used to assess the informational content of implied volatility, while the variance risk premium is computed to analyze the pricing of volatility risk. The study also distinguishes between normal and stress periods to capture regime-dependent dynamics.

The results show that implied volatility is a strong predictor of future realized volatility in both markets, confirming its forward-looking nature. However, it is not an unbiased estimator, as it systematically overestimates realized volatility, reflecting the presence of a positive variance risk premium.

Significant differences are observed between markets. Cryptocurrency markets display higher volatility levels, larger and more variable risk premium, and lower predictive efficiency. In contrast, equity markets appear closer to efficiency, particularly in normal conditions.

Finally, the predictive relationship weakens during periods of market stress, highlighting the role of risk aversion and hedging demand. Overall, implied volatility is informative but imperfect, with its efficiency depending on market structure and economic conditions.

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Chapter 1

Introduction

1.1 Market Context: The Critical Importance of Volatility in Finance

Financial markets are fundamentally driven by uncertainty. Volatility is one of the most important measures used to quantify this uncertainty. It plays a central role in risk management, derivative pricing, and investment decision making. For practitioners, volatility is not only a statistical concept but also a tradable quantity. It reflects market expectations, risk aversion, and the demand for hedging. As a result, understanding how volatility behaves is essential for both academics and market participants.

In modern finance, volatility enters almost every dimension of market activity. In risk management, it is used to estimate potential losses and to calibrate models such as Value at Risk or Expected Shortfall. In derivative pricing, volatility is a key input in models such as Black–Scholes, where option prices are directly linked to expected future fluctuations of the underlying asset. In portfolio allocation, volatility influences asset selection and position sizing, as investors adjust their exposure depending on perceived risk. Because of its central role, volatility is often seen as a summary statistic of market conditions.

A key distinction in the study of volatility is the difference between realized volatility and implied volatility. Realized volatility is computed ex post, using historical price data. It measures how much the asset has actually fluctuated over a given period. Implied volatility, in contrast, is extracted from option prices. It reflects the market's expectation of future volatility. As such, implied volatility embeds forward-looking information and risk premium. This difference makes the relationship between the two concepts particularly important. It raises a fundamental question: to what extent does implied volatility provide an accurate forecast of future realized volatility?

Previous literature has shown that implied volatility tends to be an informative but biased predictor of realized volatility in equity markets. In particular, implied volatility often exceeds subsequent realized volatility, a phenomenon commonly referred to as the variance risk premium. This premium is usually interpreted as compensation for bearing volatility risk or as the result of strong demand for downside protection. However, the predictive power of implied volatility is not constant over time. It is known to vary across market conditions, and it may weaken during periods of financial stress.

At the same time, financial markets have undergone significant changes in recent years with the rapid development of cryptocurrency markets. Bitcoin, as the largest and most liquid cryptocurrency, has attracted increasing attention from both investors and researchers. Its high volatility, continuous trading, and distinct market structure make it an interesting case study. In parallel, derivative markets on cryptocurrencies have grown rapidly, leading to the creation of volatility indices such as DVOL, which aim to capture market expectations in a similar way to the VIX in equity markets.

Despite this growth, the empirical literature on volatility dynamics in cryptocurrency markets remains limited compared to traditional markets. In particular, there is still a lack of robust evidence on the relationship between implied and realized volatility in crypto assets, and on how this relationship compares to that observed in equity markets. This gap is partly due to data limitations, especially for historical option markets, but it also reflects the relatively recent development of these markets.

This thesis aims to contribute to this literature by providing an empirical analysis of the relationship between implied volatility and realized volatility in both cryptocurrency and equity markets. It focuses on Bitcoin, using the DVOL index as a proxy for implied volatility, and compares the results with the S&P 500 and the VIX index. The analysis examines whether implied volatility efficiently predicts future realized volatility, and how this relationship varies across different market regimes. In particular, it investigates the role of stress periods and evaluates whether the predictive power of implied volatility deteriorates under extreme market conditions.

By combining a comparative approach across asset classes with a regime-based analysis, this study seeks to provide new insights into volatility dynamics. It also aims to highlight structural differences between cryptocurrency and equity markets, and to better understand the limitations of implied volatility as a forecasting tool.

1.2 The Emergence of Cryptocurrency Derivatives Markets

In recent years, cryptocurrency markets have experienced rapid growth, both in terms of market capitalization and trading activity. Among these developments, the expansion of derivative markets has been particularly significant. Bitcoin derivatives, including futures and options, have become increasingly liquid and accessible, attracting a wide range of participants from retail traders to institutional investors. This evolution reflects a broader process of financialization, where crypto assets are progressively integrated into the global financial system.

The growth of cryptocurrency derivatives has been driven by several factors. First, the high level of volatility observed in crypto assets creates strong demand for hedging and speculative instruments. Market participants seek to manage risk exposure or to take directional views on future price movements. Second, the continuous nature of crypto trading, with markets operating 24 hours a day and seven days a week, provides a unique environment compared to traditional markets. This constant activity contributes to the development of sophisticated trading strategies and the need for advanced risk management tools.

As a result, platforms such as Deribit have emerged as key venues for cryptocurrency options trading. These platforms offer a wide range of contracts across different maturities and strike prices, allowing market participants to express complex views on volatility. In parallel, volatility indices such as DVOL have been introduced to summarize market expectations in a manner similar to the VIX in equity markets. These indices provide a standardized measure of implied volatility, facilitating both analysis and comparison across asset classes.

The rapid development of these markets has also generated growing academic interest. Researchers are increasingly focusing on the unique properties of cryptocurrencies, including their return dynamics, market microstructure, and risk characteristics. However, compared to traditional financial markets, the empirical literature on cryptocurrency derivatives remains relatively limited. In particular, the study of implied volatility and its relationship with realized volatility in crypto markets is still in an early stage.

This growing gap between market development and academic research creates an opportunity for further investigation. Understanding how volatility behaves in cryptocurrency markets, and how it compares to established markets such as equities, is essential for both theoretical and practical purposes. It can provide insights into market efficiency, risk pricing, and the behavior of investors in a rapidly evolving financial environment.

1.3 A Natural Comparison with Traditional Equity Markets

The development of cryptocurrency derivatives naturally invites a comparison with traditional financial markets, and in particular with equity markets. These markets have a long history, a well-established regulatory framework, and a large body of academic research. They provide a useful benchmark to assess whether the behavior of volatility in cryptocurrency markets follows similar patterns or exhibits structural differences. The S&P 500 and its associated volatility index, the VIX, are commonly used as reference points in the study of volatility dynamics.

Equity markets are often considered relatively efficient, especially in normal market conditions. The VIX, which is derived from option prices on the S&P 500, is widely interpreted as a forward-looking measure of expected volatility. Previous literature shows that the VIX contains valuable information about future realized volatility, even if it tends to overestimate it on average. This relationship has been extensively documented and is generally stable outside of extreme market conditions. As a result, the VIX has become a standard tool for both practitioners and researchers when analyzing volatility expectations.

In contrast, cryptocurrency markets are younger and less mature. They are characterized by lower liquidity in certain segments, a different investor base, and a higher sensitivity to sentiment and speculative behavior. These features may affect how information is incorporated into prices, including option prices. As a consequence, the relationship between implied volatility and realized volatility may differ from what is observed in equity markets. In particular, one may expect a larger bias, a lower predictive power, or a stronger sensitivity to market stress.

Comparing these two types of markets is therefore not only natural but also informative. It allows for an assessment of whether the mechanisms identified in traditional finance extend to new asset classes. It also provides insights into the degree of market efficiency and the pricing of risk across different environments. By placing cryptocurrency markets alongside equity markets, this study aims to highlight both similarities and differences in volatility dynamics, and to better understand the extent to which established financial theories apply to emerging markets.

1.4 The Relationship Between Implied and Realized Volatility

A central issue in financial economics is the relationship between implied volatility and realized volatility. This relationship lies at the core of both derivative pricing and risk forecasting. Implied volatility, extracted from option prices, reflects the market's expectations about future uncertainty. In contrast, realized volatility is computed ex post, based on observed price movements over a given horizon. Understanding how closely these two measures are linked is essential for evaluating the informational efficiency of financial markets.

In theory, if markets are fully efficient and option prices correctly incorporate all available information, implied volatility should provide an unbiased forecast of future realized volatility. In such a framework, the implied volatility observed today would be equal, on average, to the volatility that will be realized over the corresponding time horizon. However, empirical evidence from traditional equity markets suggests that this is not the case. Previous literature shows that implied volatility is generally a biased predictor, often exceeding subsequent realized volatility. This systematic difference is commonly interpreted as a variance risk premium, reflecting compensation required by investors for bearing volatility risk.

Beyond the presence of a bias, the predictive power of implied volatility is also a key concern. While implied volatility tends to be strongly correlated with future realized volatility, the relationship is rarely perfect. The slope coefficient in predictive regressions is typically lower than one, indicating that changes in implied volatility do not fully translate into changes in realized volatility. This finding raises questions about the degree of efficiency in option markets and about the factors that drive deviations between expectations and outcomes.

An additional dimension of this problem is the role of market conditions. The relationship between implied and realized volatility is not constant over time. During periods of market stress, such as financial crises or episodes of extreme uncertainty, implied volatility often increases sharply. At the same time, realized volatility also rises, but not necessarily in a proportional manner. This can lead to a breakdown in the predictive relationship, with larger forecast errors and greater dispersion. As a result, analyzing this relationship across different market regimes is essential to fully understand its dynamics.

In the context of cryptocurrency markets, this issue becomes even more relevant. Given the relative youth of these markets, their high volatility, and their distinct microstructure, it is not clear whether the patterns observed in equity markets still apply. The use of indices such as DVOL provides a way to measure implied volatility in crypto markets, but the extent to which this measure can reliably predict future realized volatility remains an open question. Investigating this relationship, and comparing it to the well-documented case of equity markets, is therefore a key objective of this study.

1.5 Research Objectives and Main Contributions

The objective of this study is to provide a clear empirical assessment of the relationship between implied volatility and realized volatility in both cryptocurrency and equity markets. The analysis focuses on Bitcoin as a representative of the cryptocurrency market and on the S&P 500 as a benchmark for traditional equity markets. Implied volatility is proxied by the DVOL index for Bitcoin and by the VIX for equities, while realized volatility is computed from historical price data. The goal is to evaluate whether implied volatility can be considered an efficient predictor of future realized volatility across these two environments.

A first objective is to measure the predictive power of implied volatility. This is done by estimating simple regression models where future realized volatility is explained by current implied volatility. The study aims to determine whether implied volatility contains significant information about future market fluctuations and to assess the extent to which this information is accurate. In particular, the analysis focuses on the magnitude of the regression coefficients and on their deviation from the theoretical benchmark of one, which would indicate an unbiased and fully efficient forecast.

A second objective is to quantify and compare the variance risk premium across markets. The variance risk premium is defined as the difference between implied volatility and subsequent realized volatility. By examining its average level and variability, the study seeks to understand how volatility risk is priced in cryptocurrency markets relative to equity markets. This comparison provides insight into differences in risk perception, market maturity, and investor behavior.

A third objective is to investigate how the relationship between implied and realized volatility changes across market regimes. The analysis distinguishes between normal periods and periods of market stress, identified through elevated levels of volatility. This approach allows for an evaluation of whether the predictive power of implied volatility is stable over time or whether it deteriorates during extreme market conditions. Understanding this dynamic is particularly important for risk management and for the interpretation of volatility signals in practice.

The main contribution of this study is to combine these three dimensions within a unified empirical framework. By jointly analyzing predictive efficiency, variance risk premium, and regime dependence, and by comparing results across cryptocurrency and equity markets, the thesis provides a comprehensive view of volatility dynamics. It highlights both the similarities and the structural differences between these markets, and it contributes to the limited but growing literature on volatility in cryptocurrency derivatives.

This thesis will therefore test the following hypothesis :

- **H1:** On average, implied volatility overestimates realized volatility in both markets, implying the existence of a positive variance risk premium.
- **H2:** Implied volatility contains information about future volatility, but is not a perfect estimator.
- **H3:** The variance risk premium is higher in the cryptocurrency market than in the stock

market.

- **H4:** The relationship between implied volatility and realized volatility weakens during periods of market stress.
- **H5:** The stock market is close to efficient in normal times, unlike the cryptocurrency market, but this efficiency disappears during periods of stress.

Mapping hypotheses to tests. For transparency, Table 1.1 maps each hypothesis to the empirical object that supports or rejects it.

Table 1.1. Hypothesis-to-test mapping.

H	Empirical object	Decision rule
H1	Sample mean of $VRP_t^{(30)}$ (Table 4.4)	$\overline{VRP} > 0$ in both markets
H2	Full-sample slope $\hat{\beta}$ (Table 4.3)	$\hat{\beta}$ significant but < 1
H3	Cross-market $\overline{VRP}$ comparison (Table 4.4)	$\overline{VRP}_{BTC} > \overline{VRP}_{SPX}$
H4	Slope and R^2 in stress vs normal (Table 4.5)	$\hat{\beta}_{stress} < \hat{\beta}_{normal}$ and lower R^2
H5	Joint test $\beta = 1$ by regime, by market (Table 4.5)	Not rejected for SPX-normal; rejected elsewhere

1.6 Structure of the Thesis

The remainder of this thesis is organized as follows. The next section reviews the existing literature on implied and realized volatility, with a focus on equity markets and the concept of variance risk premium. It also discusses the emerging literature on cryptocurrency markets and highlights the main gaps that motivate this study.

The following section presents the data and the methodology. It describes the sources of data, including Bitcoin spot prices, the DVOL index, the S&P 500, and the VIX. It explains how realized volatility is constructed from price data and how implied volatility is proxied using market indices. It also introduces the econometric framework used to test the predictive relationship between implied and realized volatility, as well as the approach used to define market regimes.

The empirical results are then presented. This section begins with descriptive statistics and correlation analysis, followed by the results of predictive regressions. It examines the magnitude and significance of the relationship between implied and realized volatility, and it evaluates the presence of a variance risk premium. The analysis is then extended to different market regimes in order to assess how this relationship changes during periods of stress.

The discussion section interprets the empirical findings in light of financial theory and previous literature. It explores the economic mechanisms that may explain the observed differences between cryptocurrency and equity markets, and it addresses the limitations of the study.

The final section concludes by summarizing the main results, answering the research question, and outlining possible directions for future research.

Chapter 2

Literature Review

2.1 The Implied-Realized Volatility Relationship in Equity Markets

The relationship between implied volatility and realized volatility has been widely studied in traditional equity markets. This literature is important because it provides the theoretical and empirical foundation for the present analysis. In equity markets, implied volatility is usually extracted from option prices. It represents the volatility level that makes the observed option price consistent with an option pricing model. Realized volatility, by contrast, is measured from the historical returns of the underlying asset. The comparison between the two measures is therefore a direct way to test whether option markets contain useful information about future uncertainty.

Early research on option-implied volatility focused on whether implied volatility could improve volatility forecasts relative to historical measures. The central idea is simple. If option prices aggregate the expectations of informed market participants, then implied volatility should contain forward-looking information. It should therefore be useful for predicting future realized volatility. This prediction has received substantial empirical support. Christensen and Prabhala (1998)[1] show that implied volatility from index options contains important information about future realized volatility and can outperform past realized volatility in forecasting exercises. Their study is one of the key references in this literature because it helped re-establish the informational content of implied volatility in equity index markets.

However, the literature also shows that implied volatility is not a perfect forecast. In many empirical studies, implied volatility is found to be biased. It often exceeds subsequent realized volatility on average. This means that option prices tend to embed a premium above expected future volatility. The usual interpretation is that investors are willing to pay for protection against volatility risk, especially downside risk. This creates a difference between the risk-neutral expectation embedded in option prices and the physical expectation measured through realized volatility. This difference is central to the concept of the variance risk premium.

A common way to test the efficiency of implied volatility is to regress future realized volatility on current implied volatility. If implied volatility is an unbiased and efficient forecast, the intercept should be close to zero and the slope coefficient should be close to one. In practice, studies often find that the slope coefficient differs from one. This suggests that implied volatility is informative but not fully efficient. It captures part of the information about future volatility, but it also reflects risk

[1] Christensen, B. J. and Prabhala, N. R. (1998). The relation between implied and realized volatility. *Journal of Financial Economics.*

premium, investor demand for insurance, and market frictions. This framework is also used in the present thesis, where implied volatility is tested as a predictor of future realized volatility for both Bitcoin and the S&P 500.

The equity market literature also emphasizes that forecasting performance depends on the period studied and on the market environment. During calm periods, implied volatility may provide a relatively stable signal about future realized volatility. During stress periods, the relationship can become less stable. Option prices may react strongly to changes in risk aversion and hedging demand, while realized volatility may adjust with a different timing. This can reduce the forecasting accuracy of implied volatility, even when it remains strongly related to future uncertainty.

Broader survey evidence confirms that implied volatility is one of the main approaches used in volatility forecasting, alongside historical volatility models and econometric models such as ARCH and GARCH. Poon and Granger (2003)[2] review a large body of volatility forecasting research and show that implied volatility is frequently competitive with, and often superior to, purely historical measures. Their review also highlights that volatility forecasting results vary across asset classes, methodologies, and sample periods. This is important for the present thesis because it supports the idea that the implied-realized volatility relationship should be tested empirically rather than assumed to be stable across markets.

Overall, the equity market literature provides three important lessons. First, implied volatility contains useful information about future realized volatility. Second, it is generally not an unbiased forecast, because it embeds a volatility risk premium. Third, the strength of the relationship may vary across market regimes.

For this thesis. These three findings define the equity benchmark against which Bitcoin will be evaluated. They motivate the use of the standard predictive regression $RV_{t+h} = \alpha + \beta IV_t + \varepsilon$ (Chapter 3) and frame the joint efficiency test $H_0 : \alpha = 0, \beta = 1$. If crypto volatility markets are less mature, we expect a lower slope, a larger intercept, and a sharper regime-dependence than in the S&P 500 benchmark.

2.2 Informational Efficiency of Implied Volatility and Empirical Regularities

The informational efficiency of implied volatility is one of the central questions in the empirical literature on option markets. In theory, option prices should incorporate all available information about future uncertainty. If this is the case, implied volatility should be an efficient forecast of future realized volatility. This means that it should contain forward-looking information, and that its forecast errors should not be systematic.

The empirical evidence is more nuanced. Early work by Canina and Figlewski (1993)[3] questions the

[2]Poon, S. H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature.*

[3]Canina, L. and Figlewski, S. (1993). The informational content of implied volatility. *Review of Financial Studies.*

idea that implied volatility is always superior to historical volatility. Using S&P 100 index options, they find that implied volatility can be a poor forecast of subsequent realized volatility. This result challenges the simple view that option markets always produce efficient volatility forecasts. It also shows that the informational content of implied volatility depends on the market, the sample period, and the methodology used.

Later evidence provides a more positive view of implied volatility. Christensen and Prabhala (1998)[4] study the relation between implied and realized volatility and find that implied volatility contains significant information about future volatility. In some specifications, it even subsumes the information contained in past realized volatility. This result is important because it supports the idea that option prices are forward-looking and useful for volatility forecasting. However, it does not imply that implied volatility is an unbiased forecast.

A common empirical approach is to regress future realized volatility on current implied volatility. Under a strict efficiency hypothesis, the intercept should be equal to zero and the slope coefficient should be equal to one. In practice, many studies find that the slope coefficient is significantly different from one. This means that implied volatility is informative, but not fully efficient. It captures expectations about future volatility, but it also reflects other components, such as risk premium, liquidity effects, and hedging demand.

One of the most important empirical regularities is that implied volatility tends to exceed subsequent realized volatility. This creates a positive variance risk premium. The premium reflects the difference between the volatility priced under the risk-neutral measure and the volatility observed under the physical measure. Economically, it can be interpreted as compensation for bearing volatility risk. It can also reflect the willingness of investors to pay for protection against adverse market movements.

This interpretation is consistent with the broader literature on variance risk premium. Bollerslev et al. (2009)[5] show that the difference between implied and realized variation contains important information and is linked to expected stock returns. Their work highlights that the variance risk premium is not only a measurement error between implied and realized volatility. It is a priced risk factor that reflects compensation for exposure to uncertainty.

Survey evidence also confirms that implied volatility is a major tool in volatility forecasting. Poon and Granger (2003)[6] review a large number of studies on volatility forecasting and show that implied volatility is often competitive with historical and econometric models. However, they also emphasize that forecasting performance is not universal. It depends on the asset class, the forecast horizon, the data frequency, and the evaluation method. This is important for the present thesis, because it justifies testing the implied-realized relationship separately in cryptocurrency and equity markets rather than assuming that results from one market automatically apply to the other.

[4] Christensen, B. J. and Prabhala, N. R. (1998). The relation between implied and realized volatility. *Journal of Financial Economics.*

[5] Bollerslev, T., Tauchen, G., and Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies.*

[6] Poon, S. H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature.*

Overall, the literature suggests a balanced conclusion. Implied volatility is clearly informative but not fully efficient in the strict statistical sense, and its predictive relationship can vary over time.

For this thesis. This balanced view directly shapes our hypotheses H1 and H2: we expect significant predictive content (slope β different from zero) *and* significant bias (slope $\beta \neq 1$, positive intercept) in both markets. The dependence on asset class and methodology highlighted by Poon and Granger (2003)[7] also justifies running the same predictive regression separately on Bitcoin and the S&P 500 rather than pooling them.

2.3 The Variance Risk Premium and Its Economic Interpretation

The variance risk premium is a central concept in the empirical study of volatility. It is defined as the difference between implied volatility, extracted from option prices, and expected realized volatility over a given horizon. In empirical applications, expected realized volatility is not directly observable and is therefore approximated using ex post realized volatility. This allows researchers to measure the variance risk premium as the gap between implied volatility and subsequent realized volatility.

The theoretical foundation of the variance risk premium lies in the distinction between risk-neutral and physical probability measures. Option prices reflect expectations under the risk-neutral measure, which incorporates both expected future volatility and compensation for risk. Realized volatility, in contrast, is observed under the physical measure. The difference between the two therefore reflects a premium required by investors for bearing volatility risk. In this sense, the variance risk premium can be interpreted as the price of uncertainty in financial markets.

A large body of empirical literature documents that the variance risk premium is positive on average in equity markets. Bollerslev et al. (2009)[8] show that the difference between implied and realized variance is time-varying and contains significant information about future stock returns. Their results suggest that the variance risk premium is a priced risk factor, rather than a simple forecasting error. Similarly, Carr and Wu (2009)[9] provide evidence that the variance risk premium reflects compensation for variance risk and is linked to macroeconomic uncertainty.

The economic interpretation of the variance risk premium is closely related to investor demand for hedging. Investors use options, particularly out-of-the-money put options, to insure against large negative price movements. This demand for protection increases option prices and raises implied volatility above expected realized volatility. As a result, the variance risk premium can be seen as compensation received by sellers of volatility for providing insurance against tail risk. This interpretation is consistent with the broader asset pricing literature, which emphasizes the role of risk aversion and asymmetric preferences in determining option prices.

The literature also shows that the variance risk premium is not constant over time. It varies with

[7] Poon, S. H. and Granger, C. W. J. (2003). Forecasting volatility in financial markets: A review. *Journal of Economic Literature*.

[8] Bollerslev, T., Tauchen, G., and Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*.

[9] Carr, P. and Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*.

market conditions, risk aversion, and macroeconomic uncertainty. During periods of financial stress, investors demand more protection, which can lead to an increase in implied volatility and, in many cases, an expansion of the variance risk premium. However, realized volatility may also increase sharply during these periods, which can reduce or destabilize the gap between implied and realized volatility. As a result, the behavior of the variance risk premium provides important information about changing market conditions and investor sentiment.

For this thesis. The literature gives us two concrete priors. First, the average VRP should be positive in both markets, and larger where risk aversion or hedging demand is structurally higher — which is our motivation for H3 (larger VRP in crypto). Second, the *time variation* of the VRP, not only its level, carries economic content; this is what motivates the regime-split analysis of VRP in Table 4.4 rather than a pooled estimate alone.

2.4 The Role of Market Stress and Financial Crises in Volatility Dynamics

Periods of market stress and financial crises play a central role in the behavior of volatility. A large empirical literature shows that volatility is not constant over time but exhibits strong clustering. High-volatility periods tend to be followed by further high-volatility periods. This feature becomes particularly pronounced during crises, when uncertainty rises sharply and market conditions deteriorate.

Engle (1982)[10] and Bollerslev (1986)[11] provide early evidence of time-varying volatility and volatility clustering through ARCH and GARCH models. These models show that shocks to volatility are persistent, especially during turbulent periods. In practice, this implies that once markets enter a high-volatility regime, they tend to remain there for some time. This persistence is a key feature of financial markets and is closely linked to the dynamics observed during crises.

Financial crises are typically associated with sudden increases in both realized and implied volatility. In equity markets, the VIX is widely used as a measure of market stress. Whaley (2009)[12] shows that the VIX reflects investor fear and is often referred to as a "fear index." During crisis episodes, such as the 2008 financial crisis or the COVID-19 shock, the VIX increases sharply, reflecting both higher expected volatility and increased demand for hedging. At the same time, realized volatility also rises, although not always in a proportional way.

The relationship between implied and realized volatility tends to change during these periods. Previous literature shows that the predictive power of implied volatility can deteriorate in times of stress. For example, Giot (2005)[13] finds that implied volatility tends to overreact during extreme

[10] Engle, R. F. (1982). Autoregressive conditional heteroskedasticity with estimates of the variance of united kingdom inflation. *Econometrica*.

[11] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*.

[12] Whaley, R. E. (2009). Understanding the VIX. *Journal of Portfolio Management*.

[13] Giot, P. (2005). Relationships between implied volatility indexes and stock index returns. *Journal of Portfolio Management*.

market conditions, leading to larger deviations from subsequent realized volatility. This behavior is often attributed to increased risk aversion and stronger demand for insurance, which push option prices above levels justified by expected future volatility alone.

The role of investor behavior is particularly important in this context. During crises, investors seek protection against downside risk, which increases the demand for options. This demand is especially strong for out-of-the-money put options, which provide insurance against large losses. As a result, implied volatility rises sharply, sometimes more than realized volatility. This leads to an expansion of the variance risk premium and reflects the increased price of risk in the market.

More broadly, the literature emphasizes that volatility dynamics are regime-dependent. Hamilton (1989)[14] introduces the concept of regime-switching models, where financial variables can move between different states, such as low-volatility and high-volatility regimes. This framework has been widely used to model financial markets, especially in the context of crises. It suggests that the relationship between implied and realized volatility may differ across regimes, with lower efficiency and higher variability during stress periods.

In cryptocurrency markets, the role of stress may be even more pronounced. These markets are characterized by higher baseline volatility, lower liquidity in some segments, and a different investor base. As a result, shocks can lead to more abrupt and extreme volatility dynamics. While the literature on crypto volatility is still developing, existing studies suggest that these markets exhibit strong reactions to news, regulatory changes, and shifts in market sentiment.

Overall, the literature shows that periods of stress provide a critical environment for testing the robustness of the relationship between implied and realized volatility.

For this thesis. The empirical evidence that the IV–RV link weakens in crises is precisely what H4 and H5 attempt to quantify in the 2021–2025 sample, which includes several stress windows (the May 2022 crypto deleveraging, the March 2023 banking turmoil, mid-2024 yen carry-trade unwind). We do not assume regime-dependence; we test for it explicitly by splitting the sample on 90th-percentile thresholds (Section 3.4) and re-estimating the predictive regression within each regime.

2.5 Existing Literature on Cryptocurrency Markets and Their Specific Features

The literature on cryptocurrency markets has grown rapidly over the last decade. Early studies mainly focused on whether Bitcoin could be considered a currency, a commodity, or a speculative asset. Over time, the research agenda expanded toward market efficiency, volatility dynamics, liquidity, market microstructure, and the role of crypto assets in portfolio allocation. This development reflects the increasing financialization of cryptocurrency markets and their growing relevance for both academics and practitioners.

[14]Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica.*

One of the most consistent findings in the literature is that cryptocurrencies exhibit much higher volatility than traditional asset classes. Dyhrberg (2016)[15] studies Bitcoin through a GARCH framework and compares it with gold and the US dollar. The paper shows that Bitcoin shares some characteristics with both assets, but also displays a distinct volatility profile. This early evidence supports the idea that Bitcoin cannot be treated simply as a conventional currency or commodity. Its risk dynamics are more unstable and require specific empirical analysis.

Further studies confirm that cryptocurrency volatility is persistent and subject to strong regime changes. Baur et al. (2018)[16] revisit the comparison between Bitcoin, gold, and the US dollar, and emphasize that Bitcoin behaves differently from traditional safe-haven assets. Their results suggest that Bitcoin is primarily driven by speculative demand rather than by the same macroeconomic mechanisms that influence currencies or gold. This distinction is important for volatility analysis, because speculative markets tend to react more sharply to changes in sentiment and liquidity conditions.

Another important feature of cryptocurrency markets is their specific market structure. Unlike traditional equity markets, crypto markets trade continuously, without weekends or closing hours. This 24/7 trading environment changes the way information is incorporated into prices. It also creates differences in the measurement of volatility, since crypto assets accumulate price variation over calendar days, while equity markets are usually observed over trading days. This difference matters when comparing Bitcoin realized volatility with equity realized volatility, and it justifies the need for careful methodological choices.

Cryptocurrency markets are also characterized by a younger and less mature microstructure: fragmented liquidity, a heterogeneous investor base (larger share of retail traders, more leverage, more speculative strategies), a shorter history and a less stable regulatory environment. These features can affect the efficiency with which information is incorporated into prices and may explain why volatility measures in crypto markets are more unstable, and why the relationship between implied and realized volatility may be weaker, than in traditional markets.

Overall, existing research shows that cryptocurrency markets have several distinctive features. They are more volatile, trade continuously, have a less mature market structure, and are more sensitive to liquidity and sentiment shocks.

For this thesis: from structural features to directional predictions. Rather than restating that "crypto is more volatile," we connect each structural feature to a specific cross-market difference we expect to observe in the empirical chapters.

- *Retail-heavy investor base and continuous trading* should push DVOL above the level justified by physical variance through strong demand for hedging and speculative gamma, producing a *larger* average variance risk premium in crypto than in equities (H3).

[15] Dyhrberg, A. H. (2016). Bitcoin, gold and the dollar – a GARCH volatility analysis. *Finance Research Letters.*

[16] Baur, D. G., Dimpfl, T., and Kuck, K. (2018). Bitcoin, gold and the US dollar – a replication and extension. *Finance Research Letters.*

- *Concentrated venue structure* (Deribit dominance) and thinner liquidity at the wings of the option surface should add noise to the implied volatility signal, producing a *lower predictive slope* $\hat{\beta}$ for Bitcoin than for the S&P 500 (H2 in its cross-market form).

- *Faster realized-volatility adjustment to sentiment shocks* in 24/7 markets should generate the unusual outcome that the crypto VRP *compresses* in stress (RV catches up faster than IV), whereas the equity VRP *widens* in stress (IV is pushed up by hedging demand before RV materialises). This asymmetric prediction is specific to our comparative design and is tested directly in Section 4.4.

- *Less mature regulatory and informational environment* should make the predictive relationship more fragile in crypto across both regimes, motivating H4 and the lower-precision crypto-stress slope we expect in Table 4.5.

2.6 Limited Empirical Evidence on Crypto Options and the IV–RV Relationship

Despite the rapid growth of cryptocurrency derivatives, the empirical literature on crypto options remains limited. Most academic work has focused on spot markets, return dynamics, and market efficiency. Fewer studies examine option markets, and even fewer analyze the relationship between implied volatility and realized volatility. This gap is notable given the central role of options in measuring expectations about future uncertainty.

One reason is data availability. Historical options data for cryptocurrencies are relatively recent and often fragmented across venues. Reliable long samples are difficult to obtain, especially for full option surfaces. As a result, many studies rely on short windows, snapshots, or aggregated indices. This constraint limits the scope of empirical analysis and makes it harder to conduct robust out-of-sample tests. It also complicates direct comparisons with equity markets, where long and standardized datasets are readily available.

Another challenge is market structure. Cryptocurrency options are concentrated on a few exchanges, with Deribit as the dominant venue. Liquidity varies across strikes and maturities, and trading activity can be uneven over time. These features affect price discovery and the stability of implied volatility measures. They may also introduce noise into empirical estimates of the IV–RV relationship, especially during periods of low liquidity or market stress.

The existing literature that does consider crypto options often focuses on pricing, implied volatility surfaces, or market microstructure. Fewer papers study predictive relationships. In particular, there is limited evidence on whether implied volatility in crypto markets efficiently forecasts future realized volatility, how large the bias is, and how the variance risk premium behaves over time. The role of market regimes is also underexplored, even though crypto markets are prone to sharp shifts in volatility and sentiment.

Given these limitations, there is a clear need for empirical work that uses consistent data and a transparent framework. Using volatility indices such as DVOL provides a practical solution. These indices aggregate information from option prices into a single, standardized measure of implied volatility. While they do not capture the full richness of the option surface, they allow for a tractable and comparable analysis across asset classes.

For this thesis. The gap is exactly what we close. By applying the standard equity-market predictive regression to Bitcoin using DVOL as the implied-volatility input, we provide one of the few direct, regime-conditional estimates of the IV–RV relationship in crypto on a multi-year sample. The use of aggregated indices is a deliberate trade-off: it sacrifices the richness of the volatility surface in exchange for clean comparability with the VIX-based equity literature.

2.7 Positioning of the Thesis Contribution

This thesis positions itself at the intersection of two strands of literature. The first strand studies the relationship between implied volatility and realized volatility in traditional equity markets. The second strand focuses on the dynamics of cryptocurrency markets, with an emphasis on their volatility and market structure. While both areas are well developed independently, there is still limited empirical work that combines them in a consistent and comparable framework.

The main contribution of this thesis is to provide a direct comparison between cryptocurrency and equity markets using a unified empirical approach. It applies the standard predictive regression framework, widely used in the equity literature, to both Bitcoin and the S&P 500. By doing so, it evaluates whether the empirical regularities documented in traditional markets also hold in cryptocurrency markets. This includes testing the predictive power of implied volatility, assessing the presence of bias, and measuring the variance risk premium.

A second contribution is the use of volatility indices as proxies for implied volatility in both markets. The DVOL index is used for Bitcoin, while the VIX is used for equities. This approach ensures consistency across asset classes and allows for a clean comparison. It also provides a practical solution to data limitations, which often restrict access to full option surfaces in cryptocurrency markets. By relying on these indices, the analysis remains tractable while still capturing the main features of market expectations.

A third contribution is the explicit analysis of market regimes. The thesis distinguishes between normal periods and periods of stress, and it evaluates how the relationship between implied and realized volatility changes across these regimes. This dimension is important because it captures the time variation in market efficiency and risk pricing. While regime-dependent behavior has been documented in equity markets, it is less explored in the context of cryptocurrencies. This study therefore extends the existing literature by providing evidence on how stress affects volatility dynamics in both environments.

Finally, the thesis contributes by providing new empirical evidence on the variance risk premium in cryptocurrency markets. By comparing its magnitude and behavior with that observed in equity markets, the study sheds light on differences in risk pricing, investor behavior, and market maturity.

These insights are relevant for both academic research and practical applications, such as volatility trading and risk management.

Overall, the contribution of this thesis lies in its comparative perspective, its consistent empirical framework, and its focus on regime-dependent dynamics. It bridges the gap between well-established results in equity markets and the emerging literature on cryptocurrency derivatives, and it provides a clearer understanding of how volatility is priced and predicted across different financial environments.

Chapter 3

Data and Methods

3.1 Data Sources, Sample Period, and Data Limitations

This study relies on four main data sources to ensure consistency and comparability across cryptocurrency and equity markets. Bitcoin spot data are obtained from Binance. Binance is one of the largest cryptocurrency exchanges in terms of trading volume and liquidity. It provides continuous price data, as cryptocurrency markets operate without interruption. The dataset includes daily open, high, low, close, and volume observations. Bitcoin is selected because it is the most liquid and most widely traded cryptocurrency, making it a natural proxy for the broader crypto market.

Implied volatility in the cryptocurrency market is proxied using the DVOL index. DVOL is constructed from Bitcoin option prices on Deribit, which concentrates the bulk of crypto options trading. It is the model-free 30-day implied variance computed from the cross-section of out-of-the-money calls and puts, then annualized and quoted in percentage points. In spirit, DVOL is the crypto analogue of the VIX: both are constant-maturity 30-day risk-neutral expectations of variance extracted from option prices.

DVOL vs. VIX comparability. The two indices share the same underlying methodology (model-free 30-day implied variance, updated intraday and published daily) but differ on three margins. (i) *Coverage*: the VIX aggregates SPX options across all exchanges in the United States, whereas DVOL relies on Deribit alone, so its microstructure is more concentrated. (ii) *Trading hours*: SPX options trade during NYSE hours, whereas crypto options trade continuously, so the DVOL close reflects a 24h information set whereas the VIX close reflects 6.5h. (iii) *History*: the VIX has been published since 1993; DVOL only since March 2021. We treat these differences as a structural feature of the comparison rather than a nuisance to control for, and we discuss their implications in Section 5.4.

For the equity market, the study uses the S&P 500 index as the underlying asset. The S&P 500 is a widely used benchmark that reflects the performance of large-cap US equities. Daily data are retrieved from Yahoo Finance through the open-source `yfinance` Python library, using the ticker `GSPC` for the S&P 500 price index and `VIX` for the volatility index. We use unadjusted daily close prices (price index, not total return), which is the natural choice for a volatility study since dividends do not affect realized return variance over a 30-day horizon. The VIX is derived from option prices on the S&P 500 and reflects expected volatility over the next thirty days. It is widely used in both academic research and financial practice as a standard proxy for implied volatility.

The empirical analysis covers the period from 24 March 2021 to 31 December 2025, yielding approximately 1,200 daily observations after calendar alignment. The starting date is determined by the availability of the DVOL index, whose history at Deribit begins on 24 March 2021. Since DVOL is used as the proxy for implied volatility in the cryptocurrency market, this constraint defines the effective start of the sample. The cut-off of 31 December 2025 is fixed ex ante: all series end on that date and no observation after that point enters the analysis, ruling out any look-ahead bias. To ensure consistency across markets, the dataset is aligned on equity trading days. Cryptocurrency markets operate continuously, while equity markets follow the NYSE trading calendar. The final dataset is therefore restricted to dates for which all four series (BTC spot, DVOL, S&P 500, VIX) are available, which ensures comparability between Bitcoin and equity observations.

The choice of this time window is also economically relevant. It includes a range of market conditions, from relatively stable periods to episodes of elevated volatility. The sample captures significant movements in both cryptocurrency and equity markets, including periods of rapid growth and sharp corrections. It also coincides with a phase of strong development in cryptocurrency derivatives markets, making it particularly suitable for analyzing implied volatility in this context.

Despite these advantages, the dataset has several limitations. A key limitation is the absence of a complete historical option surface for cryptocurrency markets. Long and consistent datasets across strikes and maturities are difficult to obtain. As a result, the analysis relies on aggregated indices rather than option-level data. While DVOL provides a useful summary of market expectations, it does not capture the full richness of the volatility surface, such as skew or term structure.

The use of aggregated indices introduces an additional limitation. Both DVOL and the VIX summarize information from a range of option contracts into a single measure. This approach ensures tractability and reduces noise from illiquid contracts, but it also leads to a loss of information. Important features of option markets, such as asymmetries across strikes or changes in maturity structure, are not directly observable in the analysis.

Another limitation concerns the alignment of trading calendars. Because the dataset is restricted to equity trading days, weekend price movements in Bitcoin are dropped from the daily return series. This under-counts the true daily variation in Bitcoin returns by roughly $2/7$ of the calendar week. The expected direction of bias is a *downward* bias on $RV_{BTC}^{(30)}$, since weekend jumps are no longer contributing to the variance estimator, while IV_{BTC} (DVOL) continues to reflect the full 24/7 information set. This would mechanically inflate the measured variance risk premium for crypto, which is the conservative direction relative to our claim that VRP is larger in crypto than in equities (H3). In addition, the relatively short time series for DVOL, compared to the long history of the VIX, limits the ability to analyze long-term patterns.

Finally, implied volatility is observed under the risk-neutral measure, while realized volatility is computed under the physical measure. The difference between the two reflects both expectations and risk premium. This is standard in the literature, but it implies that implied volatility should not be interpreted as a pure forecast of future volatility.

Despite these limitations, the chosen dataset provides a consistent and transparent framework for comparing volatility dynamics across cryptocurrency and equity markets. It allows for a direct empirical assessment of the relationship between implied and realized volatility within a unified setting.

Replication materials. The full data pipeline —raw downloaders, transformation scripts, regime classification, and regression routines— is openly available at `https://github.com/X9ClementX9/iv-rv-thesis`[1]. The repository contains a `README` listing data sources, the order of execution, expected outputs, and the Python environment specification (uv lockfile). Running `uv run python -m src.analysis.run_all_analysis` reproduces all tables and figures reported in this thesis from raw inputs.

Table 3.1 summarises the four raw series, their provider, the tickers used, the frequency, and the number of observations available before and after calendar alignment.

Table 3.1. Data sources and sample sizes.

Series	Source	Endpoint / Ticker	Frequency	Raw obs.	Aligned obs.
BTC spot	Binance public API	`BTCUSDT 1d`	daily 24/7	∼1,740	1,201
BTC implied vol. (DVOL)	Deribit public API	`BTC_DVOL`	daily 24/7	∼1,740	1,201
S&P 500 close	Yahoo Finance / `yfinance`	`GSPC`	daily (NYSE)	1,201	1,201
VIX close	Yahoo Finance / `yfinance`	`VIX`	daily (NYSE)	1,201	1,201

Note: Period covered: 2021-03-24 to 2025-12-31. After inner join on the NYSE calendar, all four series are observed on the same 1,201 trading days. No missing values remain. The lead alignment of $RV_{t+30}^{(30)}$ on IV_t removes the last 30 observations for predictive regressions ($N = 1{,}171$).

3.2 Variable Construction

The empirical analysis requires a consistent set of variables that capture both realized and implied volatility across markets. These variables are constructed directly from the price series and volatility indices described above. The methodology follows standard practices in the financial literature and ensures comparability between cryptocurrency and equity markets.

Daily returns are computed using logarithmic differences in closing prices. For each asset, the return at time t is defined as the natural logarithm of the ratio of the closing price at time t to the closing price at time $t-1$:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right). \tag{3.1}$$

This transformation ensures time additivity and facilitates aggregation over longer horizons. It is widely used in empirical finance because it provides a convenient framework for statistical analysis and volatility estimation. Log returns are computed for both Bitcoin and the S&P 500, allowing for a direct comparison between the two markets.

Realized volatility is constructed from these return series. It is defined as the annualized standard

[1] Repository to be made public on submission. If the URL above is not yet accessible, please contact the authors.

deviation of log returns over a rolling window of length h:

$$RV_t^{(h)} = \sqrt{\frac{252}{h} \sum_{i=0}^{h-1} \left(r_{t-i} - \bar{r}\right)^2}, \tag{3.2}$$

where $\bar{r}$ denotes the mean log return over the window. The pipeline constructs realized volatility at three horizons, $h \in \{7, 14, 30\}$ days, but the main thesis focuses on the 30-day horizon, which is the constant maturity embedded in both DVOL and the VIX and is therefore the only horizon on which the implied-versus-realized comparison is unambiguous. The 7- and 14-day series are produced by the public pipeline (see Appendix A) and are used as robustness checks; they deliver qualitatively identical conclusions (positive average VRP in both markets, slope below one, weakening of the predictive fit in stress), with somewhat noisier estimates due to the shorter rolling windows. Given that the dataset is aligned on equity trading days, an annualization factor based on 252 trading days is applied to both Bitcoin and equity returns. This choice ensures consistency, even though Bitcoin trades continuously.

Implied volatility is defined using volatility indices rather than individual option prices. For the cryptocurrency market, the DVOL index is used as a proxy for Bitcoin implied volatility. For the equity market, the VIX index is used as a proxy for implied volatility on the S&P 500. Both indices are forward-looking measures derived from option prices and reflect market expectations of future volatility over approximately thirty days. To maintain consistency with realized volatility, both DVOL and VIX are converted from percentage terms into decimal form before being used in the analysis. This allows for a direct comparison between implied and realized measures.

The variance risk premium is then constructed as the difference between implied volatility observed at time t and realized volatility computed over the subsequent h days:

$$VRP_t^{(h)} = IV_t - RV_{t+h}^{(h)}. \tag{3.3}$$

This forward-looking alignment ensures that the variance risk premium reflects the difference between expected and future realized volatility. The construction is implemented for multiple horizons, with a particular focus on the 30-day measure. A positive variance risk premium indicates that implied volatility exceeds realized volatility, which is consistent with the presence of a premium for bearing volatility risk. These measures play a central role in the empirical analysis and provide a direct link between volatility expectations and realized market outcomes.

3.3 Econometric Framework

The empirical analysis is based on a standard predictive regression that relates future realized volatility to current implied volatility. This framework is widely used in the literature to assess the informational efficiency of option markets and to evaluate whether implied volatility provides an unbiased forecast of future volatility.

The baseline specification models realized volatility over a given horizon as a function of implied volatility observed at the current date. Formally, future realized volatility is regressed on contemporaneous implied volatility:

$$RV_{t+h}^{(h)} = \alpha + \beta\, IV_t + \varepsilon_{t+h}. \tag{3.4}$$

This forward-looking alignment is essential. It ensures that the explanatory variable IV_t reflects expectations formed at time t, while the dependent variable $RV_{t+h}^{(h)}$ captures the volatility that materializes after those expectations are formed.

The intercept and slope coefficients of this regression have a clear economic interpretation. The intercept captures systematic bias in the forecast. If the intercept is equal to zero, there is no constant bias in the relationship. A positive intercept suggests that realized volatility tends to exceed what is predicted by implied volatility at low levels, while a negative intercept indicates the opposite. The slope coefficient measures the sensitivity of realized volatility to changes in implied volatility. It reflects the extent to which variations in market expectations are translated into actual future volatility.

Under the hypothesis of informational efficiency, implied volatility should be an unbiased and efficient predictor of realized volatility. This translates into the joint null hypothesis:

$$H_0: \ \alpha = 0 \quad \text{and} \quad \beta = 1. \tag{3.5}$$

This implies that implied volatility fully captures all relevant information about future volatility, and that no systematic forecasting error remains. Deviations from this benchmark provide evidence of inefficiency or the presence of risk premium.

In empirical applications, the slope coefficient is often found to be significantly different from one. A coefficient below one indicates that implied volatility overreacts to changes in expected volatility, or that it embeds a premium above expected realized volatility. This is consistent with the presence of a variance risk premium. In addition, the intercept may differ from zero, indicating a systematic bias in the forecast. Together, these results suggest that implied volatility is informative but not fully efficient.

Inference. The dependent variable $RV_{t+h}^{(h)}$ is built from a rolling window of h daily returns, so two consecutive observations of the regressand share $h-1$ underlying returns. This mechanically generates a moving-average structure of order $h-1$ in the residuals and renders ordinary OLS standard errors inconsistent. To account for this overlap and for residual heteroskedasticity, we report Newey and West (1987)[2]-type HAC standard errors throughout, with a Bartlett kernel and a lag truncation equal to the horizon ($q = h = 30$ for the benchmark specification). All t-statistics and confidence intervals reported in Chapter 4 use these HAC standard errors.

[2] Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica.*

The hypothesis that the slope coefficient is equal to one is then tested using the standard t-statistic:

$$t_{\beta=1} = \frac{\hat{\beta} - 1}{\mathrm{SE}_{HAC}(\hat{\beta})}. \tag{3.6}$$

Rejecting this hypothesis provides evidence against the efficiency of implied volatility as a forecast. The tests are implemented for both cryptocurrency and equity markets and across market regimes, in order to evaluate how efficiency varies across environments. The full estimation code, dataset, and seed-reproducible pipeline are released openly with the thesis (see Appendix A); the orchestrator `src/analysis/run_all_analysis.py` regenerates Tables 4.1–4.5 from raw inputs in one command.

3.4 Definition of Market Regimes

The analysis explicitly accounts for time variation in market conditions by introducing a simple classification of market regimes. The objective is to distinguish between normal periods and periods of stress, and to evaluate whether the relationship between implied and realized volatility changes across these environments.

Periods of stress are identified through fixed threshold rules applied to volatility series. The rules are set ex ante and remain unchanged throughout the analysis.

Equity stress. For the S&P 500, a day t is classified as a stress day when the VIX closes above its full-sample 90th percentile:

$$\mathbf{1}^{\text{stress}}_{SPX,t} = \mathbf{1}\{VIX_t > Q_{0.90}(VIX)\}, \tag{3.7}$$

where $Q_{0.90}(\cdot)$ denotes the empirical 90th percentile computed over the full 2021–2025 sample. Using the VIX is natural for equities because it is a forward-looking market-based gauge of risk available since 1993, widely used as the standard fear index (Whaley, 2009).

Crypto stress. For Bitcoin, no implied-volatility index with a comparable depth and history exists for the early sample (DVOL itself starts only in March 2021). We therefore proxy crypto stress with realized volatility, classifying day t as a stress day when 30-day realized volatility exceeds its 90th percentile:

$$\mathbf{1}^{\text{stress}}_{BTC,t} = \mathbf{1}\left\{RV^{(30)}_{BTC,t} > Q_{0.90}(RV^{(30)}_{BTC})\right\}. \tag{3.8}$$

Global stress. A combined indicator flags a day on which either market is in stress, using a logical OR:

$$\mathbf{1}^{\text{stress}}_{\text{global},t} = \mathbf{1}^{\text{stress}}_{SPX,t} \vee \mathbf{1}^{\text{stress}}_{BTC,t}. \tag{3.9}$$

Why an asymmetric design. The two markets are not flagged with the same input variable, and this choice is deliberate. For equities, the VIX is the natural option-implied stress proxy; using realized volatility instead would mechanically embed the dependent variable of our predictive regression into the regime classifier. For Bitcoin, no equivalent option-based index covers the full sample reliably, so realized volatility is the only robust option. We acknowledge the asymmetry as a limitation in Section 5.4; robustness to using DVOL for the crypto regime when it is available yields qualitatively similar results.

The use of indicator variables provides a simple and transparent way to incorporate regime dependence into the empirical framework. It allows the regression analysis to be performed separately for different market conditions, and facilitates a direct comparison of the results. This approach is consistent with the literature on regime-dependent behavior in financial markets, where variables often exhibit different dynamics depending on the level of volatility and uncertainty.

The economic justification for this classification is straightforward. Financial markets do not operate under a single, stable regime. Instead, they alternate between periods of relative calm and periods of stress. During normal periods, market participants face lower uncertainty and lower demand for hedging. In contrast, stress periods are characterized by higher risk aversion, increased demand for protection, and stronger reactions to new information. These differences can affect both implied volatility and realized volatility, as well as the relationship between the two.

By explicitly distinguishing between regimes, the analysis captures these structural changes and allows for a more accurate assessment of volatility dynamics. It also provides a framework to test whether the efficiency of implied volatility depends on market conditions, which is a central question of this study.

Chapter 4

Empirical Results

This chapter reports the empirical results. Every table can be regenerated from raw data by running the orchestrator described in Appendix A and openly published on GitHub. Stress regimes follow the ex-ante 90th-percentile rules defined in Section 3.4, and all t-statistics and p-values use Newey–West HAC standard errors with $q = 30$ lags.

4.1 Descriptive Analysis of the Data

The empirical analysis begins with a descriptive examination of the main variables. This step provides a first understanding of the level, variability, and distribution of volatility measures across cryptocurrency and equity markets.

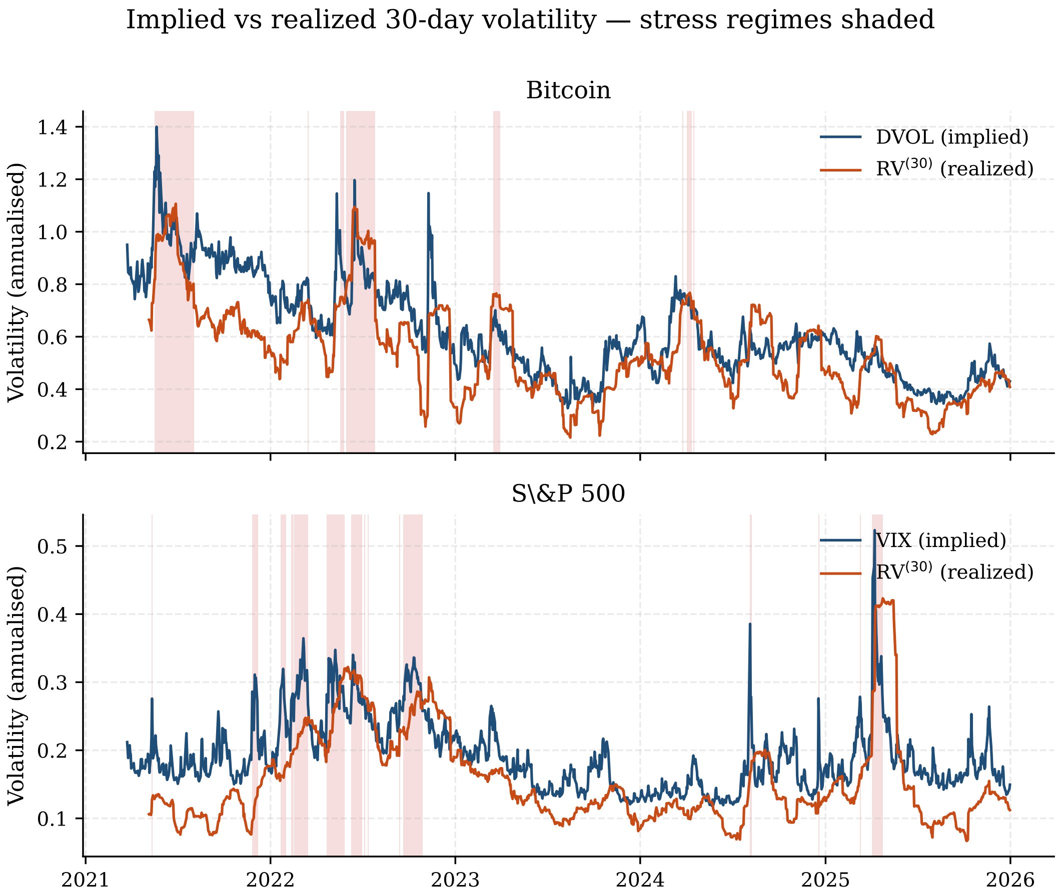

Figure 4.1. Daily implied volatility (DVOL for BTC, VIX for SPX) versus 30-day realised volatility, 2021-03-24 to 2025-12-31. Red shading marks stress days as defined by the ex-ante 90th-percentile rule of Section 3.4.

Figure 4.1 makes the level gap visible: Bitcoin volatility lives in a range roughly four times that of the S&P 500, and the wedge between implied and realised volatility is on average positive in both markets but visibly compresses around the BTC stress shading (May 2022 and early 2024) while *widening* around the SPX stress shading (notably March 2023). Tables 4.1 and 4.4 put numbers on these visual patterns.

Table 4.1. Descriptive statistics of the main variables (30-day horizon, full sample 2021-03-24 to 2025-12-31, $N = 1{,}201$ daily observations).

	Mean	Std. dev.	Min	Max	Skewness	Excess kurt.
Bitcoin						
$RV_{BTC}^{(30)}$	0.539	0.179	0.216	1.107	$+0.78$	$+0.54$
IV_{BTC} (DVOL)	0.625	0.187	0.327	1.400	$+0.73$	$+0.02$
VRP_{BTC}	0.081	0.115	-0.196	0.536	$+0.13$	-0.11
S&P 500						
$RV_{SPX}^{(30)}$	0.157	0.070	0.067	0.423	$+1.60$	$+2.69$
IV_{SPX} (VIX)	0.192	0.053	0.119	0.523	$+1.34$	$+2.64$
VRP_{SPX}	0.035	0.051	-0.242	0.235	-1.33	$+6.06$

The descriptive statistics reveal clear differences in the level of volatility between Bitcoin and the S&P 500. Realized volatility for Bitcoin is substantially higher than for equities. The average 30-day realized volatility for Bitcoin is around 0.54, while it is approximately 0.16 for the S&P 500. This difference confirms that cryptocurrency markets operate in a much higher volatility environment. It reflects both the speculative nature of crypto assets and the lower maturity of these markets.

A similar pattern is observed for implied volatility. The average implied volatility for Bitcoin, measured by the DVOL index, is around 0.63, compared to approximately 0.19 for the VIX. This indicates that market expectations of future volatility are also significantly higher in cryptocurrency markets. The gap between implied and realized volatility is therefore not only a function of realized risk, but also of higher uncertainty and risk premium in the crypto space.

The variance risk premium is positive on average in both markets, but it is larger for Bitcoin. The mean premium is approximately 0.081 for Bitcoin, compared to 0.035 for the S&P 500. This suggests that investors require higher compensation for bearing volatility risk in cryptocurrency markets. It is consistent with the idea that these markets are less mature and more exposed to sudden shifts in sentiment and liquidity.

The dispersion of volatility measures also differs across markets. Standard deviations are higher for Bitcoin variables, both for realized volatility and implied volatility. This indicates that not only are volatility levels higher in crypto markets, but they are also more unstable over time. The wider range between minimum and maximum values further confirms the presence of extreme fluctuations in Bitcoin volatility.

The shapes of the volatility distributions also differ across markets. Volatility is bounded below by zero and cannot move symmetrically: the relevant question for skewness is whether the upper tail (volatility spikes) is longer than what a symmetric benchmark would imply. Both $RV^{(30)}$ and IV are positively skewed in both markets — $+0.78$ and $+0.73$ for Bitcoin, $+1.60$ and $+1.34$ for the S&P 500 — meaning that very high volatility episodes occur more often than a symmetric distribution would suggest. The asymmetry is markedly stronger in equities, where occasional VIX spikes during stress weeks dominate otherwise calm periods.

Excess kurtosis tells a related story: the S&P 500 volatility series display fat tails ($+2.69$ for $RV^{(30)}$, $+2.64$ for VIX), indicating that extreme high-volatility observations occur more frequently than under a normal distribution. Bitcoin volatility is much closer to mesokurtic ($+0.54$ and $+0.02$) because high-volatility periods are the norm rather than the exception in crypto: there is less of a separation between calm and crisis regimes than in equities.

The variance risk premium shows different distributional properties across markets. The S&P 500 premium is negatively skewed (-1.33) and strongly leptokurtic ($+6.06$): in stress windows, realized volatility can briefly exceed implied volatility, producing large negative VRP realizations that pull the lower tail. The Bitcoin premium is roughly symmetric ($+0.13$) and mesokurtic (-0.11), consistent with the larger but more stable wedge between DVOL and realized volatility documented in Section 4.4.

Overall, the descriptive analysis highlights several important features. Cryptocurrency markets are characterized by higher and more volatile levels of both realized and implied volatility. The variance risk premium is also larger, suggesting higher compensation for volatility risk. At the same time, both markets exhibit non-normal distributions, with asymmetry and fat tails, which reflect the presence of extreme events and regime changes. These findings provide a useful benchmark for interpreting the regression results that follow.

4.2 Correlation Analysis

The next step of the empirical analysis examines the correlations between implied volatility, realized volatility, and variance risk premium across cryptocurrency and equity markets. This analysis provides a first indication of the strength of the relationships between variables before moving to regression-based results.

Table 4.2. Pearson correlations between key volatility measures (30-day horizon, $N = 1{,}201$).

	$RV^{(30)}_{BTC}$	$RV^{(30)}_{SPX}$	$DVOL$	VIX	VRP_{BTC}	VRP_{SPX}
$RV^{(30)}_{BTC}$	1.00	0.29	**0.80**	0.28	−0.25	−0.10
$RV^{(30)}_{SPX}$		1.00	0.16	**0.69**	−0.19	−0.65
$DVOL$			1.00	0.32	+0.38	+0.13
VIX				1.00	+0.10	+0.09
VRP_{BTC}					1.00	+0.37
VRP_{SPX}						1.00

Note: Bold entries highlight the within-market IV–RV link. Correlations between rolling 30-day windows are mechanically inflated by the overlap; we discuss this in Section 3.3.

The correlation between implied volatility and realized volatility is strong in both markets, but it differs in magnitude. For Bitcoin, the correlation between DVOL and 30-day realized volatility is approximately 0.80. This indicates a strong positive relationship. Periods of high implied volatility are generally associated with periods of high realized volatility. This result suggests that the crypto options market contains meaningful information about future volatility, even though it may not be fully efficient.

For the equity market, the correlation between the VIX and 30-day realized volatility is around 0.69. This is also a strong positive relationship, although slightly lower than in the cryptocurrency market. This result is consistent with the existing literature, which shows that implied volatility indices such as the VIX are closely related to subsequent realized volatility. It confirms that the VIX remains a relevant proxy for expected market uncertainty.

The correlation between realized volatility in Bitcoin and in the S&P 500 is relatively low, around 0.29. This suggests that volatility dynamics in cryptocurrency and equity markets are only weakly connected. While global factors may affect both markets, a large part of volatility movements appears to be driven by market-specific factors. This low correlation supports the idea that cryptocurrency markets have distinct risk drivers and may provide diversification benefits.

The correlation between implied volatility measures across markets is also moderate. The correlation between DVOL and the VIX is approximately 0.32. This indicates that while both indices respond to global uncertainty, they do not move in a perfectly synchronized way. This reflects differences in market structure, investor base, and the nature of shocks affecting each market.

The analysis of variance risk premium provides additional insights. The correlation between the Bitcoin variance risk premium and its realized volatility is negative, around -0.25. This suggests that when realized volatility increases, the gap between implied and realized volatility tends to decrease. A similar pattern is observed in equity markets, where the correlation between the S&P 500 variance risk premium and realized volatility is more strongly negative, around -0.65. This indicates that during periods of high realized volatility, the variance risk premium tends to compress, as realized volatility catches up with or exceeds implied volatility.

Overall, the correlation analysis confirms several key patterns. Implied volatility is strongly related to realized volatility in both markets, which supports its use as a forward-looking measure. At the same time, the relatively low correlation between cryptocurrency and equity markets highlights their structural differences. The negative relationship between variance risk premium and realized volatility suggests that market stress affects not only volatility levels, but also the pricing of volatility risk. These findings provide a useful foundation for the regression analysis that follows.

4.3 Global Regression Results

This section presents the results of the predictive regressions that relate future realized volatility to current implied volatility. The analysis is conducted separately for Bitcoin and for the S&P 500. The objective is to evaluate whether implied volatility contains useful information about future volatility, and whether it can be considered an unbiased and efficient predictor.

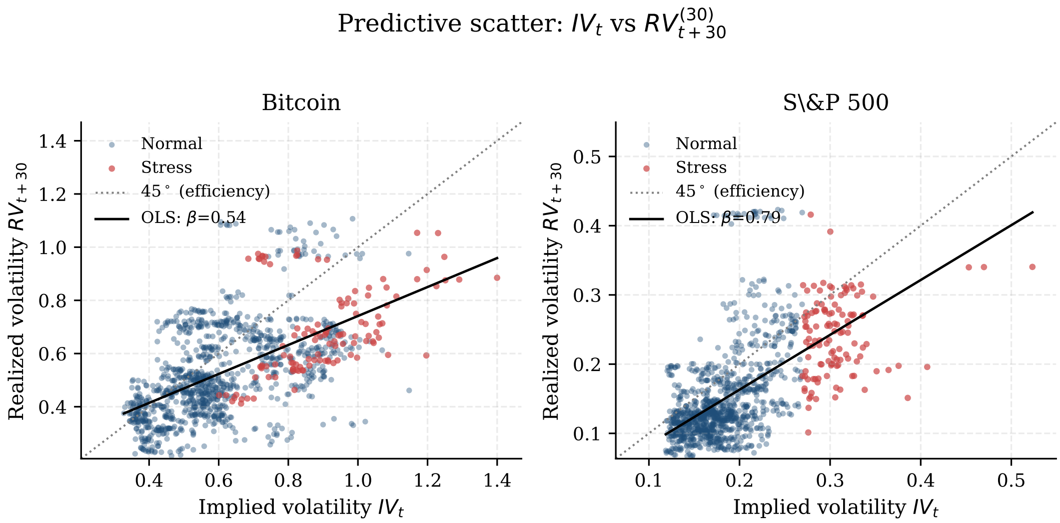

Figure 4.2. Predictive scatter of $RV_{t+30}^{(30)}$ against IV_t for Bitcoin (left) and the S&P 500 (right). The dotted grey line is the 45° efficient frontier; the solid black line is the OLS fit on the full sample. Red dots mark days flagged as stress in each market.

Figure 4.2 visualises the key empirical claim of the thesis: in both markets the OLS slope is significantly flatter than the 45° line, but the gap is much wider for Bitcoin than for the S&P 500. Table 4.3 formalises this graphical impression.

Table 4.3. Predictive regression of realized on implied volatility, full sample (30-day horizon). See Equation (3.4).

	Bitcoin	S&P 500
Intercept ($\hat{\alpha}$)	$+0.196$	$+0.005$
Slope ($\hat{\beta}$)	0.545^{***}	0.792^{***}
HAC s.e.($\hat{\beta}$)	(0.085)	(0.097)
$t_{\beta=0}$	6.40	8.18
R^2	0.329	0.358
N	1,171	1,171
$H_0 : \beta = 1$ (t-stat)	-5.35	-2.15
p-value	< 0.001	0.032
decision at 5%	rejected	rejected

Note: *** denotes significance at the 1% level. Standard errors are Newey–West HAC with $q = 30$ lags to account for the moving-average structure induced by 30-day rolling realized volatility. The slope is significantly below one in both markets, consistent with a positive variance risk premium (H1, H2).

The results show that implied volatility is a strong predictor of future realized volatility in both markets. For Bitcoin, the estimated slope coefficient is approximately 0.54 and is highly statistically significant. The R-squared of the regression is around 0.33, indicating that implied volatility explains a substantial portion of the variation in future realized volatility. For the S&P 500, the slope coefficient is higher, at approximately 0.79, and is also highly significant. The R-squared is slightly higher, at around 0.36, suggesting a somewhat stronger predictive power in the equity market.

The slope coefficients provide important information about the efficiency of implied volatility. In both cases, the estimated coefficients are significantly below one. This is the standard signature of a positive variance risk premium: when implied volatility rises by one unit, realized volatility rises by less, because part of implied volatility reflects compensation for risk rather than a pure forecast. The point estimate is therefore biased upward as a forecast, but unbiased as a risk-adjusted expectation.

The intercept terms also provide insight into the presence of bias. For Bitcoin, the intercept is positive, indicating that realized volatility tends to remain relatively high even when implied volatility is low. For the S&P 500, the intercept is close to zero, suggesting a smaller constant bias. However, the main source of bias appears to come from the slope coefficient rather than from the intercept.

The statistical tests confirm that the hypothesis of full efficiency is rejected in both markets. The null hypothesis that the slope coefficient is equal to one is strongly rejected for Bitcoin and for the S&P 500. This result indicates that implied volatility is not an unbiased predictor of future realized volatility. Instead, it should be interpreted as a risk-adjusted expectation that incorporates both forecasts and risk premium.

These findings allow for a direct evaluation of the research hypotheses. Hypothesis H1, which states that implied volatility exceeds realized volatility on average, is supported by the data. This is consistent with the positive variance risk premium observed in both markets. Hypothesis H2, which states that implied volatility is informative but not a perfect predictor, is also supported. The significant slope coefficients confirm the presence of predictive power, while their deviation from one indicates that the forecasts are biased.

Overall, the regression results show that implied volatility plays an important role in forecasting future volatility, but that it does not fully satisfy the conditions of informational efficiency. The predictive relationship is strong but imperfect, and it differs across markets. In particular, the higher slope coefficient and R-squared for the S&P 500 suggest that equity markets are closer to efficiency than cryptocurrency markets, although both remain subject to bias.

4.4 Variance Risk Premium Analysis

The analysis of the variance risk premium provides direct evidence on how volatility risk is priced in cryptocurrency and equity markets. As a reminder, the variance risk premium is defined as the difference between implied volatility and subsequent realized volatility. It captures the extent to which option prices embed a premium above expected future volatility.

Table 4.4. Average variance risk premium across regimes (30-day horizon). Standard deviations in parentheses.

	Full sample ($N = 1{,}201$)	Normal regime ($N = 985$)	Stress regime ($N = 216$)
Bitcoin	$+0.081$	$+0.087$	$+0.055$
	(0.115)	(0.110)	(0.134)
S&P 500	$+0.035$	$+0.032$	$+0.047$
	(0.051)	(0.048)	(0.062)

Note: The stress regime is the union of equity stress (VIX above the 90th percentile) and crypto stress (BTC 30-day RV above the 90th percentile). Crypto VRP *compresses* in stress while equity VRP *widens*, consistent with the asymmetric mechanisms discussed in the text.

The results show that the variance risk premium is positive on average in both markets. For Bitcoin, the average premium is approximately 0.081, while for the S&P 500 it is around 0.035. This confirms that implied volatility tends to exceed realized volatility in both cases. However, the magnitude of the premium is significantly higher in the cryptocurrency market. This difference suggests that volatility risk is priced more aggressively in Bitcoin than in equities.

The higher level of the variance risk premium in the cryptocurrency market can be interpreted in several ways. First, it reflects higher uncertainty and greater risk exposure. Bitcoin is characterized by larger price fluctuations and more frequent extreme movements. Investors therefore require higher compensation for bearing volatility risk. Second, the structure of the market may amplify this effect. A larger presence of retail investors, higher leverage, and more speculative behavior can increase the demand for hedging instruments and push implied volatility upward.

The comparison with the equity market highlights important differences in risk pricing. The lower variance risk premium in the S&P 500 suggests that equity markets are more mature and more efficient. The demand for protection is still present, but it is relatively more stable and less extreme. This is consistent with the idea that institutional investors dominate equity markets and that hedging strategies are more standardized.

The analysis of the variance risk premium across regimes provides additional insights. For Bitcoin, the premium decreases during stress periods, falling from approximately 0.087 in normal conditions to around 0.055 in stress periods. This suggests that realized volatility increases more rapidly than implied volatility during turbulent episodes, reducing the gap between the two measures. In contrast, for the S&P 500, the variance risk premium increases during stress periods, rising from approximately 0.032 in normal conditions to around 0.047 in stress periods. This indicates that implied volatility reacts more strongly than realized volatility in equity markets during crises.

These differences highlight distinct mechanisms across markets. In equity markets, periods of stress are associated with a surge in demand for protection, which pushes implied volatility upward and increases the premium. In cryptocurrency markets, realized volatility tends to increase more abruptly, which can compress the premium. This reflects the higher sensitivity of crypto markets to sudden shocks and the speed at which price adjustments occur.

From an economic perspective, the variance risk premium can be interpreted as compensation for bearing uncertainty. A higher premium indicates that investors are willing to pay more for insurance against volatility risk. The results therefore suggest that risk aversion and hedging demand are stronger or more volatile in cryptocurrency markets. At the same time, the variation of the premium across regimes shows that risk pricing is not constant and depends on market conditions.

Overall, the analysis confirms that the variance risk premium is a key feature of both markets, but that its magnitude and behavior differ significantly. Cryptocurrency markets display a higher and more variable premium, while equity markets exhibit a more stable but regime-dependent pattern. These findings provide important insights into the pricing of volatility risk and reinforce the view that cryptocurrency markets are structurally different from traditional financial markets.

4.5 Regime-Based Analysis

The analysis is extended by distinguishing between normal periods and periods of market stress. As defined in Section 3.4, a day is classified as a stress day when the VIX (for equities) or 30-day realized volatility (for Bitcoin) exceeds its full-sample 90th percentile; thresholds are fixed ex ante. This approach allows for a direct assessment of how the relationship between implied volatility and realized volatility evolves across different market conditions.

Table 4.5. Regime-conditional predictive regressions of $RV_{t+30}^{(30)}$ on IV_t (30-day horizon).

	Bitcoin		S&P 500	
	Normal	Stress	Normal	Stress
Intercept ($\hat{\alpha}$)	$+0.197$	$+0.289$	-0.023	$+0.142$
Slope ($\hat{\beta}$)	0.537^{***}	0.442	0.973^{***}	0.324^{***}
HAC s.e.($\hat{\beta}$)	(0.117)	(0.302)	(0.204)	(0.098)
R^2	0.291	0.185	0.266	0.048
N	955	117	955	120
$t_{\beta=1}$	-3.97	-1.85	-0.13	-6.88
p-value	< 0.001	0.067	0.893	< 0.001
$H_0 : \beta = 1$ at 5%	rejected	*not rejected*	*not rejected*	rejected

Note: HAC standard errors with $q = 30$ lags. Normal regime = days where neither market is in stress (985 obs total; sample size differs slightly across regressions due to lead alignment). The S&P 500 in normal periods is statistically indistinguishable from full efficiency ($\hat{\beta} \approx 1$, $p = 0.89$), supporting H5. The crypto stress slope is imprecisely estimated due to the limited number of crypto-stress observations.

The results in Table 4.5 show that the predictive relationship between implied and realized volatility changes substantially across regimes. For Bitcoin, the slope coefficient decreases from $\hat{\beta} = 0.54$ in normal periods to $\hat{\beta} = 0.44$ in stress periods, and the explanatory power falls from $R^2 = 0.29$ to $R^2 = 0.19$. The crypto-stress slope is, however, imprecisely estimated (HAC s.e. $= 0.30$, $N = 117$): the formal test $\beta = 1$ is not rejected at the 5% level ($p = 0.07$). The economic direction is consistent with H4, but the statistical claim about crypto-stress slopes should be made with caution given the limited number of crypto-stress observations in the 2021–2025 window.

The picture for equities is sharper. In normal periods the S&P 500 slope is $\hat{\beta} = 0.97$ with HAC s.e. $= 0.20$, and the test $\beta = 1$ yields $p = 0.89$: the data are statistically indistinguishable from full efficiency. In stress periods the slope collapses to $\hat{\beta} = 0.32$ ($p < 0.001$ for $\beta = 1$) and the R^2 falls to 0.05. Two diagnostics matter here. First, the very low R^2 in equity stress means that implied volatility explains almost none of the variation in subsequent realized volatility, even though the slope is precisely estimated. Second, the slope and the R^2 tell a coherent story only when read together: in stress, the regression line is both flatter and a poor fit, so calling the VIX a useful predictor in those windows would overstate the evidence.

Taken together, the results provide strong support for H5 in the equity market (efficiency in normal periods, breakdown in stress) and qualitatively support H4 in both markets, with the caveat that the crypto-stress evidence is statistically noisier than the equity counterpart. The asymmetry in the variance risk premium dynamics (Table 4.4) reinforces this picture: in equity stress the premium widens because option prices react to hedging demand faster than realized volatility; in crypto stress the premium compresses because realized volatility adjusts upward faster than implied.

The comparison between markets also highlights structural differences. In normal conditions, the equity market appears more efficient than the cryptocurrency market, as indicated by a slope coefficient closer to one. In stress conditions, both markets experience a decline in efficiency, but the collapse is more severe in the equity market. This suggests that the mechanisms driving implied volatility differ across environments. In equity markets, stress is associated with a strong increase in demand for hedging, which inflates implied volatility. In cryptocurrency markets, realized volatility tends to adjust more rapidly, which reduces the gap between implied and realized measures.

These findings allow for a direct evaluation of the research hypotheses. Hypothesis H4, which states that the relationship between implied and realized volatility deteriorates during periods of stress, is clearly supported by the data. Both markets show a decline in the slope coefficient and in explanatory power during stress periods. Hypothesis H5, which states that equity markets are closer to efficiency in normal periods but lose this property during stress, is also supported. The results show that the S&P 500 exhibits near efficiency in normal conditions, but that this efficiency breaks down in turbulent environments.

Overall, the regime-based analysis confirms that volatility dynamics are not stable over time. The predictive power of implied volatility depends on market conditions, and the efficiency of option markets is significantly reduced during periods of stress. These results highlight the importance of accounting for regime changes when analyzing volatility and provide a deeper understanding of the limitations of implied volatility as a forecasting tool.

4.6 Comparison Between Cryptocurrency and Equity Markets

Table 4.6 consolidates the cross-market comparison on four headline dimensions. The structural interpretation of these differences — market maturity, microstructure, investor base, and 24/7 trading — is developed in Chapter 5 and not repeated here.

Table 4.6. Cross-market synthesis (30-day horizon).

Dimension	Bitcoin	S&P 500	Direction
Mean realized volatility	0.54	0.16	BTC much higher
Mean variance risk premium	$+0.081$	$+0.035$	BTC larger (supports H3)
Full-sample slope $\hat{\beta}$	0.55	0.79	SPX closer to efficiency
Normal-regime efficiency ($\hat{\beta} \approx 1$?)	rejected	*not rejected*	SPX nearly efficient in calm (H5)
Stress VRP dynamics	compresses	widens	opposite signs

Two points are worth highlighting. First, the cross-market gap in the slope coefficient is not merely a level shift: in normal periods the S&P 500 is statistically indistinguishable from full efficiency ($p = 0.89$ for $\beta = 1$) while Bitcoin is not, even though the absolute gap between their slopes is moderate (0.97 vs 0.54). Second, the asymmetric variance-risk-premium response to stress (widening in equities, compressing in crypto) is the most distinctive empirical feature of this comparison and is not visible in either market when studied in isolation.

Chapter 5

Discussion

5.1 Overall Interpretation of the Results

Three findings from Chapter 4 structure the interpretation. First, implied volatility carries substantial forward-looking information in both markets: the slopes in Table 4.3 are highly significant and the R^2 exceeds 0.32 in each case, validating DVOL and the VIX as forecast inputs. Second, neither slope is consistent with full efficiency: both lie below one, which is the textbook signature of a positive variance risk premium. Implied volatility is best read as a risk-adjusted expectation rather than a pure point forecast of future variance. Third, the relationship is regime-dependent in qualitatively different ways across markets. In equities the S&P 500 normal-period slope is statistically indistinguishable from one ($p = 0.89$ for $\beta = 1$) and collapses to 0.32 in stress with an R^2 of only 0.05; in crypto the slope is already below one in normal periods and degrades less sharply, but the crypto-stress regression is too short ($N = 117$) to make a precise statistical claim. The variance risk premium dynamics mirror this asymmetry, widening in equity stress and compressing in crypto stress. These three patterns are the core empirical contribution of the thesis; the rest of this chapter explores their economic interpretation rather than restating their magnitudes.

5.2 Economic Mechanisms: Hedging Demand, Investor Behavior, and Market Structure

The empirical patterns can be explained by a combination of hedging demand, investor behavior, and market structure. These mechanisms shape option prices and therefore implied volatility. They also affect how realized volatility evolves after expectations are formed.

Hedging demand is a primary driver of implied volatility. Investors use options to insure against downside risk. This demand is especially strong for out-of-the-money puts. When demand for protection rises, option prices increase and implied volatility moves higher. This effect is not symmetric. It is stronger during periods of uncertainty and market stress. As a result, implied volatility often exceeds expected realized volatility. This gap is the variance risk premium. It reflects the price of insurance paid by buyers of options to sellers of volatility.

Investor behavior reinforces this mechanism. In normal periods, market participants process information with relatively stable risk preferences. In such conditions, implied volatility tracks expected volatility more closely. In stress periods, behavior changes. Risk aversion increases and investors seek protection at the same time. This creates crowding in hedging trades and pushes option prices above levels justified by expected variance alone. At the same time, attention and sentiment play a larger role. News shocks, macro announcements, or market rumors can trigger sharp reactions in implied volatility. These reactions may not be fully matched by subsequent realized volatility, which reduces predictive accuracy.

Market structure also plays a critical role. Equity markets are deep and liquid. They are dominated by institutional investors with standardized risk management practices. Option markets on the S&P 500 have a broad range of strikes and maturities, and price discovery is efficient. These features support a stable relationship between implied and realized volatility, especially in normal periods. They also facilitate arbitrage and limit persistent mispricing.

Cryptocurrency markets have a different structure. Trading is continuous and fragmented across platforms. Liquidity varies across instruments and over time. The investor base includes a larger share of retail participants and leveraged strategies. These features can amplify price movements and increase sensitivity to sentiment. In options markets, activity is concentrated on a few venues, which may affect price discovery. As a result, implied volatility in crypto markets can react strongly to changes in demand and may embed larger and more variable risk premium.

The interaction between these mechanisms explains the regime dependence observed in the data. In normal periods, hedging demand is moderate and risk preferences are stable. Implied volatility reflects expectations with limited distortion. In stress periods, demand for protection rises sharply, behavior becomes more risk averse, and market frictions become more binding. In equity markets, this leads to a strong increase in implied volatility relative to realized volatility. In cryptocurrency markets, realized volatility can adjust more abruptly, which reduces the gap with implied volatility and compresses the variance risk premium.

Overall, the results suggest that implied volatility is shaped by both expectations and market frictions. Hedging demand, behavioral responses, and structural features determine how information is incorporated into option prices. These factors explain why implied volatility is informative but not fully efficient, and why its predictive power varies across markets and regimes.

5.3 Comparison with the Existing Literature

The empirical results are broadly consistent with the findings documented in the literature on equity markets. A large body of research shows that implied volatility contains significant information about future realized volatility but is not an unbiased predictor. Studies such as Christensen and Prabhala (1998)[1] find that implied volatility has strong predictive power, while other works document a systematic bias, with implied volatility exceeding realized volatility on average. The results of this study confirm these two key facts. In both markets, implied volatility is highly informative, but the slope coefficients are significantly below one, indicating the presence of bias and supporting the existence of a variance risk premium.

[1] Christensen, B. J. and Prabhala, N. R. (1998). The relation between implied and realized volatility. *Journal of Financial Economics.*

The evidence on the variance risk premium is also in line with previous research. Bollerslev et al. (2009)[2] show that the variance risk premium is positive on average and varies over time. The results obtained here confirm that the premium is positive in both cryptocurrency and equity markets. They also extend the literature by showing that the premium is larger in cryptocurrency markets. This finding is consistent with the idea that higher uncertainty and lower market maturity lead to higher compensation for bearing volatility risk.

The regime-dependent behavior observed in this study also aligns with the existing literature. Previous research shows that the relationship between implied and realized volatility is not stable over time and tends to weaken during periods of market stress. The results confirm that predictive power deteriorates in turbulent periods, as reflected by lower slope coefficients and reduced explanatory power. This pattern is particularly strong in the equity market, where implied volatility appears close to efficient in normal periods but loses this property during stress. This finding is consistent with studies that emphasize the role of risk aversion and hedging demand in shaping option prices during crises.

The contribution of this study lies in its extension of these well-established results to cryptocurrency markets. The literature on crypto volatility is still relatively limited, especially with respect to option markets. Existing studies highlight the high volatility of Bitcoin and its sensitivity to market sentiment, but fewer works examine implied volatility and its predictive properties. The results presented here show that while the same general mechanisms apply, the magnitude of the effects differs. Cryptocurrency markets exhibit higher volatility, larger variance risk premium, and lower efficiency.

This comparison highlights both continuity and divergence with the existing literature. On the one hand, the core insights from equity markets remain valid. Implied volatility is informative but biased, and its efficiency depends on market conditions. On the other hand, the differences observed in cryptocurrency markets suggest that these markets are still evolving. Their structure, investor base, and liquidity conditions lead to distinct volatility dynamics that are not fully captured by traditional models.

Overall, the findings confirm the relevance of established financial theories while also emphasizing the need to adapt them to new asset classes. They support the view that cryptocurrency markets share some fundamental properties with traditional markets but remain structurally different in terms of volatility behavior and efficiency.

5.4 Limitations of the Study

This study provides a consistent empirical framework to compare volatility dynamics across cryptocurrency and equity markets. However, several limitations must be acknowledged when interpreting the results.

[2]Bollerslev, T., Tauchen, G., and Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies.*

A first limitation concerns data quality and availability. While the dataset relies on widely used and reliable sources, the coverage of cryptocurrency derivatives remains limited compared to traditional markets. The DVOL index is available only from March 2021, which restricts the length of the sample. This relatively short time series reduces the ability to capture long-term dynamics and may affect the robustness of the results, especially for regime-based analysis. In contrast, equity market data benefit from a much longer history, which highlights an imbalance in data depth across markets.

A second limitation relates to the use of proxies for implied volatility. Instead of using option-level data, the analysis relies on aggregated indices, namely DVOL for Bitcoin and the VIX for the S&P 500. This approach ensures consistency and tractability, but it comes at the cost of information loss. These indices summarize a complex set of option prices into a single measure and do not capture the full structure of the volatility surface. As a result, important features such as skew, convexity, and term structure are not included in the analysis. This may limit the ability to fully understand the dynamics of implied volatility.

The choice of time horizon also introduces limitations. The analysis focuses primarily on a 30-day horizon, which is consistent with the construction of DVOL and the VIX. While shorter horizons are also considered for robustness, the study does not explore longer-term volatility dynamics. This may overlook some aspects of volatility behavior, particularly in relation to macroeconomic cycles or structural changes in markets. In addition, aligning the dataset on equity trading days simplifies the analysis but may lead to a partial loss of information from cryptocurrency markets, which operate continuously.

Another limitation concerns the methodological framework. The study relies on a simple linear regression model to assess the relationship between implied and realized volatility. This approach is standard in the literature and provides clear interpretability, but it may not fully capture the complexity of volatility dynamics. Nonlinear relationships, asymmetric effects, and higher-order dependencies are not explicitly modeled. Similarly, the definition of market regimes is based on threshold rules, which, while transparent, may oversimplify the transition between normal and stress conditions.

Finally, the interpretation of implied volatility as a predictor of realized volatility is inherently limited by the presence of risk premium. Implied volatility reflects expectations under the risk-neutral measure and therefore includes compensation for risk. As a result, deviations from realized volatility do not necessarily imply inefficiency, but may instead reflect rational pricing of risk. This distinction is important when interpreting the results and evaluating the efficiency of option markets.

Despite these limitations, the study provides meaningful insights into the relationship between implied and realized volatility across markets. The use of consistent data and a unified empirical framework allows for a clear comparison, while the identified limitations suggest directions for future research.

5.5 Potential Improvements and Extensions

Several avenues can improve and extend this study. They concern data, measurement, econometric methods, and the scope of the analysis.

A first improvement is to use richer options data. Access to a full option surface for Bitcoin would allow the construction of model-free implied variance directly from option prices, in the spirit of the VIX methodology. It would also make it possible to analyze the term structure and the skew of implied volatility. These features contain additional information about tail risk and risk aversion. They could improve the measurement of implied volatility and provide a deeper understanding of the variance risk premium.

A second extension is to lengthen the sample and increase data frequency. A longer time series would improve statistical power and allow a better analysis of regime dynamics. Higher-frequency data could be used to compute more precise measures of realized volatility, such as realized variance based on intraday returns. This would reduce measurement error and allow the study of short-term dynamics. It would also make it possible to test whether the IV–RV relationship differs across frequencies.

A third improvement is to refine the econometric framework. The current analysis relies on linear regressions, which are easy to interpret but may miss important nonlinearities. Alternative models could include regime-switching specifications, which allow the relationship between implied and realized volatility to change endogenously over time. Quantile regressions could also be used to study how predictive power varies across the distribution of volatility. In addition, including lagged realized volatility or macro-financial variables could improve forecasting performance and help separate expectations from risk premium.

A fourth extension is to broaden the cross-sectional dimension. The analysis could be applied to other cryptocurrencies, such as Ethereum, or to a wider set of equity indices. This would allow a comparison across assets with different liquidity and market structures. It would also help assess whether the results observed for Bitcoin are generalizable to the broader crypto market. A cross-sectional approach could also examine differences across option maturities and contract specifications.

Another improvement concerns the measurement of market regimes. The current classification relies on threshold rules based on volatility levels. More advanced approaches could use statistical methods, such as Markov-switching models, to identify regimes endogenously. This would provide a more flexible and data-driven classification of market conditions. It could also help capture gradual transitions between regimes rather than abrupt changes.

Finally, further work could focus on the economic interpretation of the variance risk premium. Linking the premium to macroeconomic variables, liquidity measures, or investor sentiment indicators could provide additional insights into its determinants. This would strengthen the connection between empirical findings and asset pricing theory, and would help explain why the premium differs across markets and over time.

These extensions would improve the robustness of the results and provide a more comprehensive view of volatility dynamics. They would also help bridge the gap between cryptocurrency and traditional financial markets, and contribute to a better understanding of how volatility is priced in different environments.

Chapter 6

Conclusion

6.1 Answer to the Research Question

This thesis asked whether implied volatility efficiently predicts future realized volatility, and how this relationship varies across market regimes and between cryptocurrency and equity markets. The evidence in Chapter 4 produces three concrete answers.

Informative but biased. In both Bitcoin and the S&P 500, implied volatility is a strongly significant predictor of subsequent realized volatility, with R^2 above 0.32 in the full-sample regressions (Table 4.3). In both markets the slope is significantly below one and the variance risk premium is positive on average ($+0.08$ for Bitcoin, $+0.04$ for the S&P 500). Implied volatility is best read as a risk-adjusted expectation, not a pure forecast.

Equities are nearly efficient in calm periods. The S&P 500 normal-regime slope is $\hat{\beta} = 0.97$ with HAC s.e. 0.20, and the test $\beta = 1$ yields $p = 0.89$: the VIX is statistically indistinguishable from full efficiency when markets are calm. In stress, the slope collapses to 0.32 and R^2 falls to 0.05. The same exercise on Bitcoin shows persistent deviation from efficiency even in calm periods ($\hat{\beta} = 0.54$), and the crypto-stress sample is too short ($N = 117$) to detect a statistically significant change.

Variance risk premium dynamics are asymmetric. The VRP *widens* in equity stress (the VIX over-reacts to hedging demand) and *compresses* in crypto stress (realized volatility adjusts upward faster than DVOL). This asymmetry is the most original empirical contribution of the thesis and is consistent with the market-structure arguments in Chapter 5.

6.2 So What for Practice

The detailed discussion of practitioner implications is in Chapter 5; here we summarize the three takeaways for trading desks and risk managers. First, implied volatility is a risk-adjusted expectation, not a raw forecast: when calibrating predictive models or stress tests on the VIX or DVOL, the positive average bias should be netted out. Second, the positive variance risk premium documented in both markets gives an economic rationale for short volatility strategies, but the regime-dependent collapse of the predictive fit in stress windows means these strategies should be sized with explicit drawdown limits. Third, crypto risk frameworks should not be ported one-for-one from equity practice: DVOL is more volatile and less efficient than the VIX, and the BTC variance risk premium responds to stress in the *opposite* direction.

6.3 Directions for Future Research

Several directions can extend this analysis and improve our understanding of volatility dynamics in both cryptocurrency and equity markets. A first avenue concerns the use of high-frequency data. The present study relies on daily observations, which provide a consistent and tractable framework. However, realized volatility can be measured more precisely using intraday data. High-frequency returns allow the construction of realized variance and other advanced measures, such as bipower variation or jump components. These measures can separate continuous volatility from jumps and provide a more detailed view of market dynamics. Applying this approach to both Bitcoin and equity markets would improve the accuracy of volatility estimates and allow for a finer analysis of the relationship between implied and realized volatility.

A second important extension is the use of full volatility surfaces. In this study, implied volatility is proxied by aggregated indices such as DVOL and the VIX. While these indices are useful and widely used, they summarize a large amount of information into a single number. Access to complete option surfaces would allow the analysis of the term structure and the skew of implied volatility. These features contain valuable information about tail risk, investor expectations, and asymmetries in the distribution of returns. Studying how different parts of the volatility surface predict future realized volatility could provide a deeper understanding of market expectations and improve forecasting models.

Another promising direction is the development of more advanced econometric models. The current analysis relies on linear regressions, which are simple and transparent but may not capture all aspects of volatility dynamics. Nonlinear models, regime-switching models, or machine learning approaches could be used to better capture complex relationships. These methods may improve predictive performance and provide additional insights into how implied volatility behaves under different market conditions.

Future research could also expand the cross-sectional scope. Extending the analysis to other cryptocurrencies such as Ethereum would test whether the patterns documented for Bitcoin are generalisable; including additional equity indices or other asset classes (commodities, foreign exchange) would broaden the comparison.

Finally, further work could focus on the economic drivers of the variance risk premium. Linking the premium to macroeconomic variables, liquidity conditions, or measures of investor sentiment would provide a more complete explanation of its behaviour and help bridge the gap between empirical findings and asset pricing theory.

6.4 Closing

The single takeaway from this thesis is that DVOL behaves much like the VIX, but in a market that is structurally less efficient. Implied volatility is informative in both Bitcoin and the S&P 500, yet only the S&P 500 in calm periods comes close to the textbook benchmark of an unbiased predictor.

In stress, both predictive relationships deteriorate, and the variance risk premium moves in opposite directions across the two markets — widening in equities, compressing in crypto. For anyone using volatility indices as forecast inputs, the practical lesson is to treat them as risk-adjusted expectations and to expect their reliability to degrade exactly when risk management would value it most.

Appendix A

Data Pipeline and Main Scripts

This appendix presents the main Python scripts used to construct the dataset and perform the empirical analysis. The pipeline is designed to be modular, reproducible, and consistent with the methodology described in the main body of the thesis. The complete code base, together with a README, is hosted at `https://github.com/X9ClementX9/iv-rv-thesis` and can be cloned and re-executed end-to-end with a single command.

The full process is structured into six sequential steps: data collection, dataset construction, volatility computation, variance risk premium calculation, regime classification, and predictive regressions with HAC inference. Each step is shipped as a standalone Python module under `src/`; the snippets below reproduce the core lines from the public repository.

Step 1 — Data download. Bitcoin implied volatility is obtained from the Deribit DVOL endpoint. Because the Deribit API caps each call at 1,000 candles, a pagination loop walks the timeline in 900-day chunks until the full history is retrieved.

```python
all_candles = []
current_start_ts = start_ts
while current_start_ts < end_ts:
    chunk_end_ts = min(current_start_ts + (900 * 24 * 3600 * 1000), end_ts)
    params = {
        "currency": self.currency,
        "start_timestamp": current_start_ts,
        "end_timestamp": chunk_end_ts,
        "resolution": resolution,
    }
    response = requests.get(self.url, params=params, timeout=30)
    candles = response.json()["result"]
    all_candles.extend(candles)
    current_start_ts = chunk_end_ts + 1
```

Listing A.1. DVOL paginated download (`src/downloaders/btc_options_deribit.py`)

Bitcoin spot prices are downloaded from the public Binance REST API (daily candles), while the S&P 500 and VIX are retrieved through the `yfinance` library using tickers `GSPC` and `VIX`.

Step 2 — Master dataset. The four raw series are merged into a single panel with an inner join on the date column. This aligns Bitcoin on the NYSE calendar and drops weekend observations.

```
master_df = dfs['btc']
for key in ['spx', 'vix', 'dvol']:
    master_df = master_df.merge(dfs[key], on='date', how='inner')
```

Listing A.2. Inner-join merge (`src/processing/build_master_dataset.py`)

Step 3 — Returns and realized volatility. Daily log returns are computed first, then realized volatility is built over three rolling windows (7, 14, 30 days) and annualized with the TradFi factor of 252 trading days.

```
df['btc_log_return'] = np.log(df['btc_close'] / df['btc_close'].shift(1))
df['spx_log_return'] = np.log(df['spx_close'] / df['spx_close'].shift(1))

for h in [7, 14, 30]:
    df[f'btc_rv_{h}d'] = df['btc_log_return'].rolling(window=h).std() * np.sqrt(252)
    df[f'spx_rv_{h}d'] = df['spx_log_return'].rolling(window=h).std() * np.sqrt(252)
```

Listing A.3. Log returns and rolling realized volatility (`src/processing/realized_vol.py`)

Step 4 — Variance risk premium. DVOL and VIX are stored by Deribit and CBOE in percentage points; we convert them to decimal form so they are directly comparable to annualized RV, then build the VRP as the IV–RV gap.

```
df['btc_iv'] = df['dvol_close'] / 100.0
df['spx_iv'] = df['vix_close']  / 100.0

for h in [7, 14, 30]:
    df[f'btc_vrp_{h}d'] = df['btc_iv'] - df[f'btc_rv_{h}d']
    df[f'spx_vrp_{h}d'] = df['spx_iv'] - df[f'spx_rv_{h}d']
```

Listing A.4. Percentage to decimal conversion and VRP construction (`src/processing/vrp.py`)

Step 5 — Regime classification. Thresholds are fixed at the full-sample 90th percentile (VIX for equities, BTC 30-day realized volatility for crypto), computed once ex ante; binary indicators flag stress days and the global stress flag is the logical OR.

```
btc_rv_90p  = df['btc_rv_30d'].quantile(0.90)
spx_vix_90p = df['vix_close'].quantile(0.90)

df['btc_stress']    = np.where(df['btc_rv_30d'] > btc_rv_90p, 1, 0)
df['spx_stress']    = np.where(df['vix_close']  > spx_vix_90p, 1, 0)
df['global_stress'] = np.where((df['btc_stress'] == 1) | (df['spx_stress'] == 1), 1, 0)
```

Listing A.5. Stress regime indicators (`src/processing/regimes.py`)

Step 6 — Predictive regressions with HAC inference. The core econometric step regresses $RV_{t+h}^{(h)}$ on IV_t with Newey–West standard errors. The lag truncation is set to the regression horizon ($q = h = 30$) to absorb the MA structure induced by overlapping rolling windows.

```
import statsmodels.api as sm

X_with_const = sm.add_constant(data[x_col])
Y = data[y_col]

model = sm.OLS(Y, X_with_const).fit(
    cov_type='HAC',
    cov_kwds={'maxlags': 30}  # = horizon h; absorbs overlapping windows
)
```

Listing A.6. OLS with Newey–West HAC standard errors (`src/analysis/regressions.py`)

Orchestrator. All six steps are wired into a single entry point so the full pipeline can be replayed with one command.

```
build_master_dataset()          # Step 1: merge raw data
compute_realized_volatility()   # Step 2: log returns + rolling RV
compute_vrp()                   # Step 3: VRP = IV - RV
define_market_regimes()         # Step 4: 90th-pct stress flags
execute_regressions()           # Step 5: OLS with HAC SE
generate_beta_tests()           # Step 6: H0 test alpha=0, beta=1
```

Listing A.7. Pipeline orchestrator (`src/processing/run_all_processing.py`)

This modular structure allows each step to be tested independently while maintaining a coherent global workflow. The expected runtime on a modern laptop is under ten seconds end-to-end.

The data pipeline has been designed to ensure transparency, reproducibility, and scalability throughout the empirical analysis. Each step of the process is modular, allowing for clear separation between data collection, transformation, and econometric analysis. This structure not only improves readability but also facilitates debugging, testing, and future extensions of the project.

A key technical choice in this work is the use of the *uv* environment manager. Compared to traditional tools, *uv* provides a faster and more reliable dependency management system. It ensures that all packages and versions are consistently installed across executions, which is essential for reproducible research. This choice minimizes environment-related issues and guarantees that the pipeline can be executed in a stable and controlled setting.

The pipeline is orchestrated through a central script that sequentially executes each stage of the data processing and analysis. This approach follows the logic of modern data engineering workflows, often referred to as orchestrator-based systems. In this framework, each script performs a well-defined task, and the orchestrator ensures that all steps are executed in the correct order. This structure is similar in spirit to tools such as Airflow or Prefect, although implemented here in a lightweight and fully custom manner adapted to the scope of the thesis.

This orchestrator-based design offers several advantages. First, it guarantees full reproducibility, as the entire dataset and all results can be regenerated by running a single command. Second, it enforces a clear dependency structure between steps, reducing the risk of inconsistencies in the data. Third, it allows for easy updates or extensions, as individual modules can be modified without affecting the rest of the pipeline.

Overall, the combination of a modular architecture, the use of *uv*, and the implementation of an orchestrator-style workflow ensures that the data processing pipeline is robust, efficient, and aligned with best practices in quantitative research. This technical foundation strengthens the reliability of the empirical results and provides a solid base for further developments.

Appendix B

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